

Lab-5: Finite difference for ODE in Python

The aim of this lab is to apply the Finite Difference Method to find approximate solutions to Ordinary Differential Equations (ODE).

Instructions:

- Clone your repository from the UPPA gitlab: it contains the class `Mesh1D` for the mesh and the beginning of the first exercise.
- Do one .py file per exercise and add it to the deposit.
- The figures plotted must also be added to your git repository.
- It is recommended that you spend approximately three weeks on this assignment, as two further assignment sheets will follow.

Exercises:

Exercise 1. Discrete Laplacian in 1D with Dirichlet boundary condition.

We consider the Laplace equation with Dirichlet boundary condition:

$$\begin{cases} -u''(x) = f(x), & \forall x \in [0; 1], \\ u(0) = 0, \\ u(1) = 0. \end{cases} \quad (5.1)$$

The finite difference method leads to determine $u_h = (u_i)_{0 \leq i \leq n}$ approximating $u\left(\frac{i}{n}\right)$ when $n \rightarrow +\infty$ by solving system

$$\begin{cases} \frac{-u_{i-1} + 2u_i - u_{i+1}}{h^2} = f_i, & 1 \leq i \leq n-1 \\ u_0 = 0, \\ u_n = 0, \end{cases} \quad (5.2)$$

where $h = \frac{1}{n}$ and $f_i = f\left(\frac{i}{n}\right)$. Then, the domain $[0; 1]$ is discretized with n intervals (i.e. $n+1$ points).

In the rest of this exercise, the function f will be given by $f(x) = \sin(x)$.

1. Calculate on paper the theoretical order of the numerical scheme. Compute the exact solution u_{exact} of (5.1).
2. Write (on paper) the previous system as

$$A (u_j)_{0 \leq j \leq n} = (b_j)_{0 \leq j \leq n}$$

where the unknown is the vector $b \in \mathbb{R}^{n+1}$ and $A \in \mathcal{M}_{n+1}(\mathbb{R})$ is a matrix that we will specify.

3. Use your class `Mesh1D` to build the mesh.

4. Write a function `initLaplacianDirichlet(mesh)` returning the matrix A and a function `initb(mesh)` depending on the `mesh` and returning b .
5. Compute the numerical solution $(u_i)_{0 \leq i \leq n}$ of (5.2) with $n = 10$. Plot numerical solution obtained on $[0; 1]$. Add the exact solution on the graph.
6. For different values of n , varying from $n = 2^2$ to $n = 4096 = 2^{12}$, we will construct a graph of the error $\|u_h - u_{\text{exact}}\|_{L^\infty}$ as a function of $h = 1/n$ in log-log scale. What is the numerical order of your finite difference method?
7. For different values of n , varying from $n = 2^2$ to $n = 4096 = 2^{12}$, we will construct a graph of the condition number of the matrix A as a function of $h = 1/n$ in log-log scale. How does the condition number of A evolve as a function of h .
8. Optional: do the same by using the preconditioner of Jacobi. Interpret the results.
9. Optional: do the same by using the preconditioner SSOR. Interpret the results.

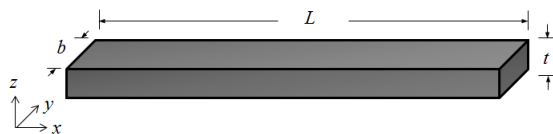
Exercise 2. We consider the following differential equation

$$\begin{cases} u''(x) = u(x) + \frac{2}{3}e^x, & \forall x \in [0; 1], \\ u(0) = 0, \\ u(1) = e/3 \end{cases} \quad (5.3)$$

1. Write the second order finite difference numerical scheme and the associated linear system of size $n + 1$. Check the theoretical order of the scheme.
2. Compute the exact solution of (5.3).
3. Write functions `initA` and `initb` depending on the mesh `mesh` and returning the matrix A and the second member b of the linear system $A u = b$ where the unknown is $u = (u_0, u_1, \dots, u_n)^T$.
4. Compute the numerical solution $(u_i)_{0 \leq i \leq n}$ of (5.2) with $n = 10$. Plot numerical solution obtained on $[0; 1]$. Add the exact solution on the graph.
5. For different values of n , varying from $n = 2^2$ to $n = 4096 = 2^{12}$, we will construct a graph of the error $\|u_h - u_{\text{exact}}\|_{L^\infty}$ as a function of $h = 1/n$ in log-log scale. What is the numerical order of your finite difference method?

Exercise 3. Static beam equation

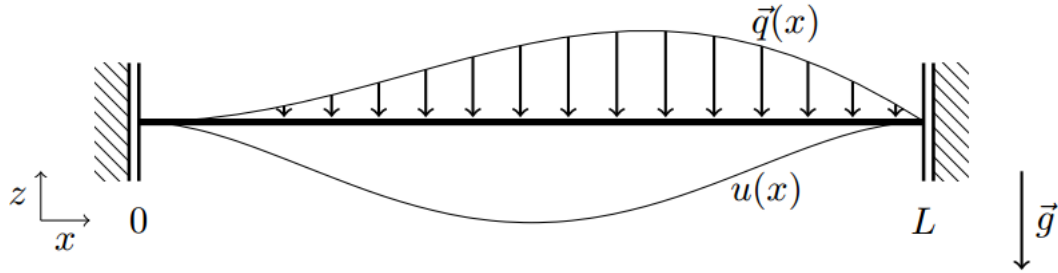
Part 1 : Linear elasticity problem: We are interested in the deformations of a beam under the effect of its weight and a load. The system consists of a beam of length L , width b and thickness t :



Several assumptions are used to simplify the problem:

- the material is homogeneous and isotropic
- the plane sections remain plane during the deformation;
- Bernoulli's hypothesis: the plane sections of the beam remain perpendicular to the mean curve during the deformation;

In this context, we consider only a 1D beam, and $u(x)$ represents the displacement in the z direction at a point x .



The vector displacement u is a solution of the Euler-Bernoulli equation, with boundary conditions for a fixed beam on the left and on the right:

$$\begin{cases} EIu^{(4)}(x) = f(x), & x \in [0; L] \\ u(0) = u'(0) = 0 \\ u(L) = u'(L) = 0 \end{cases} \quad (5.4)$$

where the constants E is the Young's modulus of the material and I is the area of the moment of inertia. The second member f is the total load of the beam including its own weight: $f(x) = -(dbt + q(x))g$, where d is the density of the material and g is the gravity constant.

Writing the numerical scheme Write the second order finite difference scheme for this problem, with a discretization of $n + 1$ points:

- Write the scheme for the points inside the domain u_i for $i = 2, \dots, n - 2$. We will compute a second order approximation of the fourth derivative.
- Write the equations for the points u_1 and u_{n-1} to be consistent with the equation and the boundary conditions.
- Give the linear system to solve, including the boundary points.

The calculations are to be given in the report.

Hint: for the boundary conditions we look for the coefficients of a linear combination of $u(x - h)$, $u(x + h)$, $u(x + 2h)$, $u(x + 3h)$ and $u(x + 4h)$ whose limited development in 0 is identified with the expression of the boundary condition: $au_{-1} + bu_1 + cu_2 + du_3 + eu_4 = hu'(0) + O(h^6)$. We write then u_{-1} in terms of the remainder and replace $u'(0)$ by its value. It only remains to replace in the numerical scheme for u_1 .

Implementation Implement the scheme with the following numerical values which correspond to a wooden beam and a zero load $q(x) = 0$:

- $L = 2m$
- $b = 30cm$

- $t = 3\text{cm}$
- $d = 480\text{kg/m}^3$
- $g = 9.81\text{m/s}^2$
- $E = 1.3 \times 10^{10}\text{N/m}^2$
- $I = bt^3/12$

We will take $n = 20$ points.

1. Verify that the function :

$$u(x) = \frac{-dbtg}{24EI}x^2(L-x)^2$$

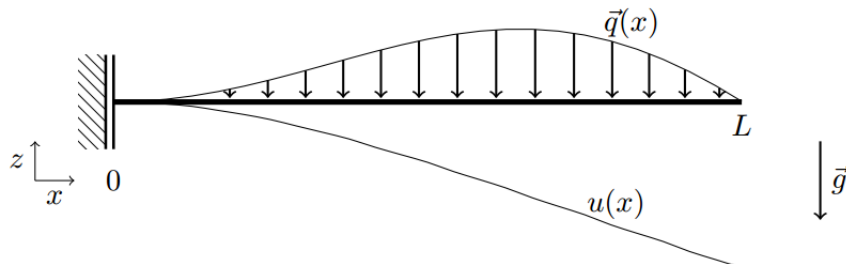
is the solution of the problem when $q(x) = 0$.

2. Implement the numerical scheme for solving the problem (5.4).
 - Write a function `initA(mesh)` allowing to assemble the coefficients of the matrix A obtained previously. The lines of the matrix corresponding to the points that involve the boundary conditions will be specified separately from the points inside the domain.
 - Write a function `initb(mesh)` allowing to initialize the second member of the system. The second member contains explicitly the values of the boundary conditions of Dirichlet type boundary conditions (here for $i = 0$ and $i = n$). For the other points, we will compute the values as a function of $f(x_i)$.
3. Plot the numerical solution and the exact solution as a function of x .
4. Plot the L^∞ -error as a function of L/n in order to numerically validate the order of the numerical scheme. What do you obtain? Why?
5. **Optional:** now use a derivative of order two (instead of order 5) to impose the condition $u'(0) = 0$ and plot the L^∞ -error as a function of L/n . What order do you obtain?
6. Compute the conditioning number of the problem matrix for some values of n .
7. Add to the model an object of mass 70kg and length 20cm placed on the beam at 30cm from the right edge of the beam. Compute the function $q(x)$ allowing to model this system. Add this solution to the graph of Question 3.

All curves are to be commented and included in the report.

Part 2: Free beam on the right

In this second part, the beam is cantilevered: it is only fixed on its left side.



$$\begin{cases} EIu^{(4)}(x) = f(x), & x \in [0; L] \\ u(0) = u'(0) = 0 \\ u''(L) = u'''(L) = 0 \end{cases} \quad (5.5)$$

Write the numerical scheme for the condition in $x = L$ for the points x_{n-1} and x_n .

1. As in the previous part, we will combine developments to order 6 to express $u(x+h)$ as a function of $u(x)$, $u(x-h)$, $u(x-2h)$ and $u(x-3h)$ to form the approximation of the equation $u^{(4)}(x_{n-1}) = h^4 f(x_{n-1})/EI$ using $u'''(L) = 0$.

Hint: check that $17u(L+h) = 40u(L) - 30u(L-h) + 8u(L-2h) - u(L-3h)$.

2. Similarly, compute an approximation to order 6 of $u(x+2h)$ as a function of $u(x)$, $u(x-h)$, $u(x-2h)$ and $u(x-3h)$. Combine this expression with that of $u(x+h)$ from the previous question to form the approximation of the equation $u^{(4)}(x_n) = h^4 f(x_n)/EI$. $u''(L) = 0$.

Hint: check that $17u(L+2h) \simeq 130u(L) - 208u(L-h) + 111u(L-2h) - 16u(L-3h)$

3. Give the system to solve.
4. Solve this new system numerically with and without the weight of the question 1.5. Plot these two solutions.
5. We wish to decrease the displacement of the end of the beam. Is it better to increase the thickness t or the width b of the end of the beam? Justify with the equation and validate with your program.

The calculations, the curves and their comments are to be included in the report.